

In-Vehicle Coupon Recommendation

Data Analysis and
Predictive Modeling

Assignment On

**In-Vehicle Coupon Recommendation: Data Analysis and
Predictive Modeling**

Course: Business Analytics (K301)

Submitted to

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Background

The **In-Vehicle Coupon Recommendation** dataset captures consumer decision-making in dynamic driving scenarios influenced by destination urgency, passenger type, weather, and available coupon categories. It models how situational factors interact with individual characteristics to shape responses to promotional offers. These insights enable businesses to design personalized, context-aware marketing strategies that boost coupon redemption and enhance customer satisfaction.

This dataset (UCI ID: 603) is publicly available through the UCI Machine Learning Repository, with raw data at “<https://archive.ics.uci.edu/static/public/603/data.csv>”. Created in 2017, it studies whether a person will accept a coupon under different driving scenarios. Data was collected via an **Amazon Mechanical Turk survey**, where participants responded to simulated driving situations. No preprocessing was applied before publication.

Designed for classification tasks in the business domain, the dataset is multivariate with **12,684 instances** and **26 features**. The target variable, Y, indicates **coupon acceptance** (1) or rejection (0). The dataset contains a mix of categorical and integer features, encompassing demographics (gender, age, marital status, education, occupation, income), scenario-specific attributes (destination, time, weather, passenger type, driving distance), and consumer habits (frequency of visits to bars, coffee houses, take-away outlets, and restaurants). Several features contain missing values. In the dataset, coupon_5 and coupon_15 means *driving distance to the restaurant/bar for using the coupon* is within 5 and 15 minutes respectively. And coupon_25 means the driving requires more than 25 minutes.

This dataset is valuable for marketing and consumer behavior analysis, helping businesses optimize promotional strategies, tailor offers, and improve engagement. It is also suitable for developing interpretable classification models, as discussed in the introductory paper: “*A Bayesian framework for learning rule sets for interpretable classification*” (Wang et al., 2017, JMLR, Journal of Machine Learning Research).

EDA and Descriptive Statistics

Dataset Structure and Feature Overview (Please refer to Appendix A)

Statistical Summary of Numerical Features

	has_children	coupon_5	coupon_15	coupon_25	direction_same	direction_opp	accepted_coupon?
count	12684.00	12684.0	12684.00	12684.00	12684.00	12684.00	12684.00
mean	0.41	1.0	0.56	0.12	0.21	0.79	0.57
std	0.49	0.0	0.50	0.32	0.41	0.41	0.50
min	0.00	1.0	0.00	0.00	0.00	0.00	0.00
25%	0.00	1.0	0.00	0.00	0.00	1.00	0.00
50%	0.00	1.0	1.00	0.00	0.00	1.00	1.00
75%	1.00	1.0	1.00	0.00	0.00	1.00	1.00
max	1.00	1.0	1.00	1.00	1.00	1.00	1.00

Table 1: Statistical Summary of Numerical Features

Univariate Analyses

Coupon acceptance rate is higher while travelling in opposite direction to the restaurant/shop, than while travelling in the same direction.

As this feature shows a strong correlation with the target, we dropped the column to avoid biased modelling.

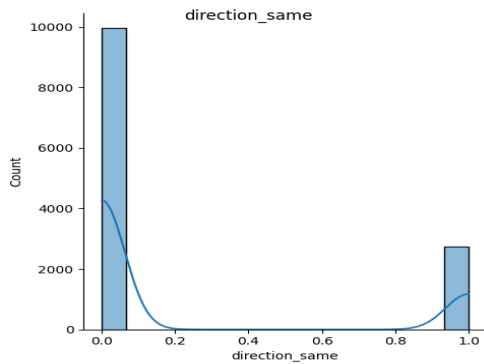


Figure 1: Acceptance when direction_same

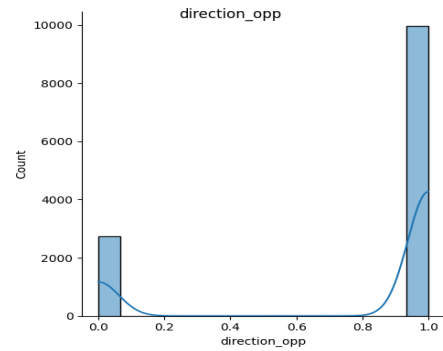


Figure 2: Acceptance when direction_opposite

Bivariate and Multivariate Analyses

Afternoon has the least amount of acceptance rates and during afternoon, there's the highest acceptance rate of the coupon category 'take-out' and 'restaurants with expenditure <\$20'.

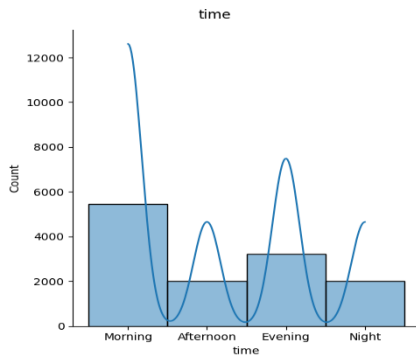


Figure 4: Acceptance by time

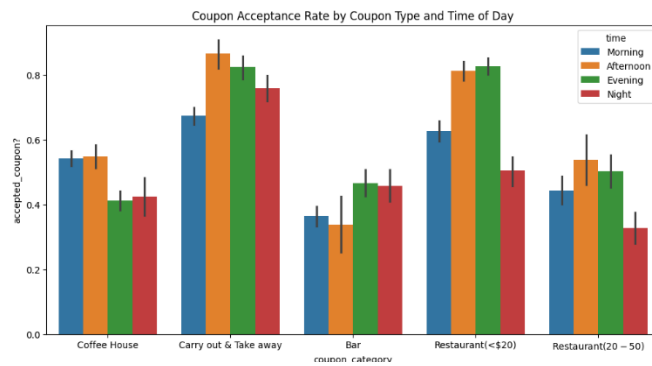


Figure 3: Coupon Acceptance Rate by Coupon Type and Time

Coupon acceptance rate is one of the highest among divorced and single males who have children.

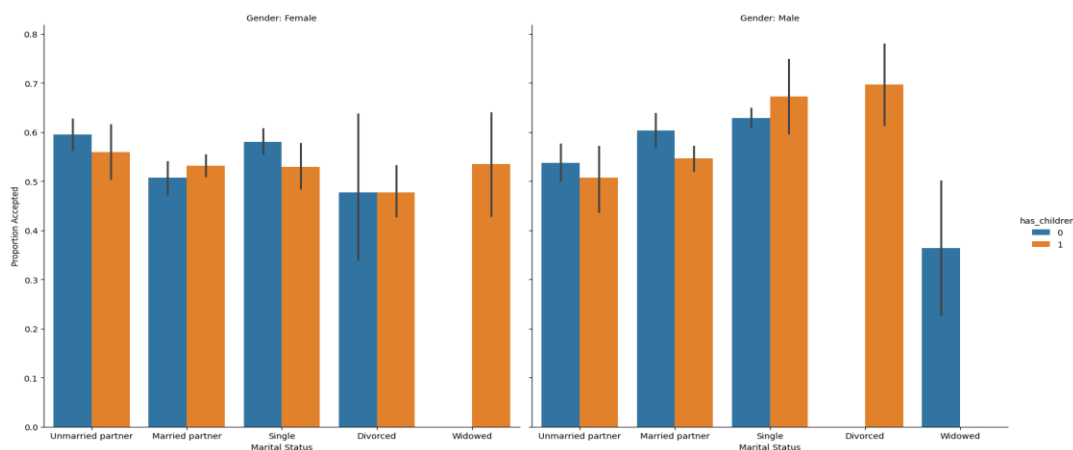


Figure 5: Acceptance Rate by Marital Status and Gender

When the expiry limit is 1 day instead of 2 hours, coupon acceptance rate is 22% higher for restaurants with expenditure \$20-\$50, 24% higher for restaurants with expenditure less than \$20. So, these two segments should focus more on offering coupons with 1 day expiry.

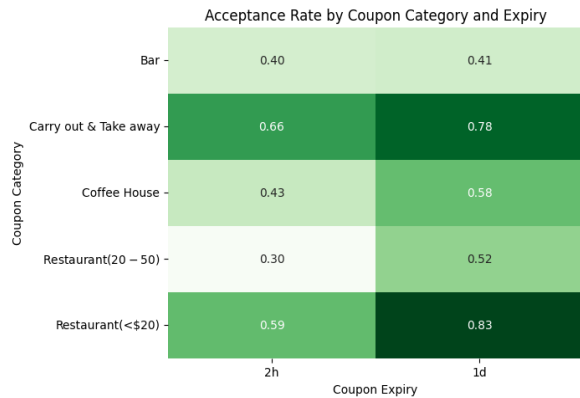


Figure 6: Acceptance Rate by Income and Coupon_5

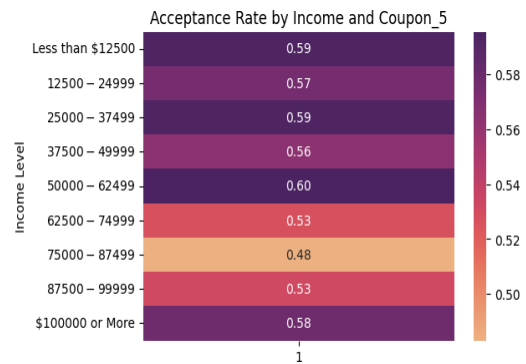


Figure 7: Acceptance Rate by Coupon_25

Marketing efforts to offer coupons should be reduced to customer within the income range of \$62,500-\$99,999, as their acceptance rate is significantly lower than others, irrespective of driving distance.

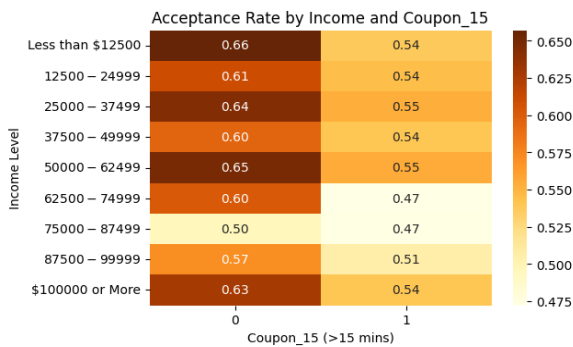


Figure 9: Acceptance Rate by Income and Coupon_15

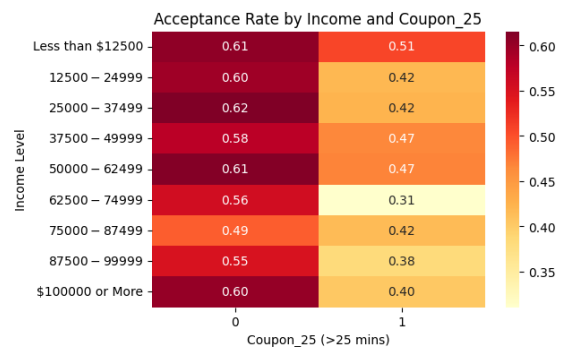


Figure 10: Acceptance Rate by Income and Coupon_25

Female employees of Architectue & Engineering field are 86% likely to accept coupons, followed by females in Construction & Extraction (77%) and Protective Services (75%). In contrast, highest acceptance rate among the male employees is 70% and that's in the field of Personal Care Services.

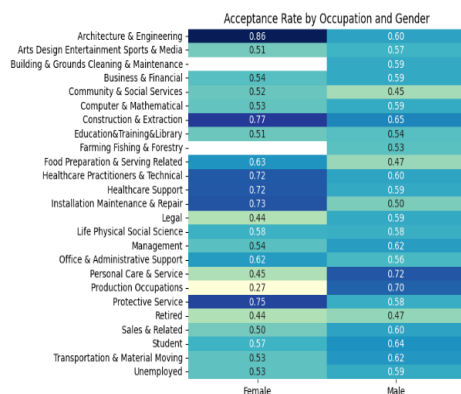


Figure 11: Acceptance Rate by Occupation & Gender

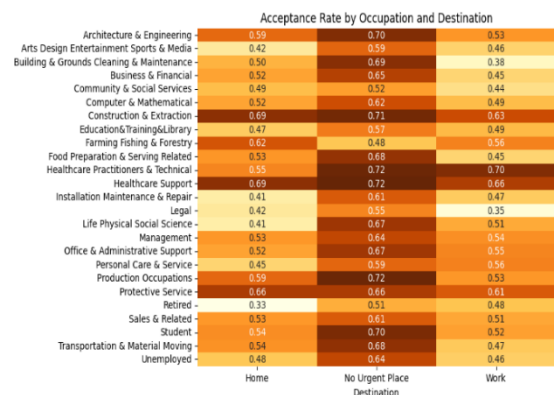


Figure 12: Acceptance Rate by Occupation & Destination

Healthcare Practitioners are most likely to avail discount coupons while going to work whereas employees of the fields Construction & Extraction, Healthcare Support and Protective Services are most likely to avail such coupons while going home from work. Almost people from all job fields likes to avail these coupons while they've no urgent work, with the highest likelihood of acceptance among Healthcare Practitioners and people from Healthcare Support and Production Occupations. Employees of Healthcare Support, Construction and Extraction, and Protective Services are very inclined to avail discount coupons irrespective of their destinations.

Besides, male and female both are more likely to accept coupons when they are alone in contrast to when they have any passengers with them. (Please refer to Appendix D for the visuals)

Coupon acceptance rate is almost equal for discount in bars, irrespective of expiry limits.

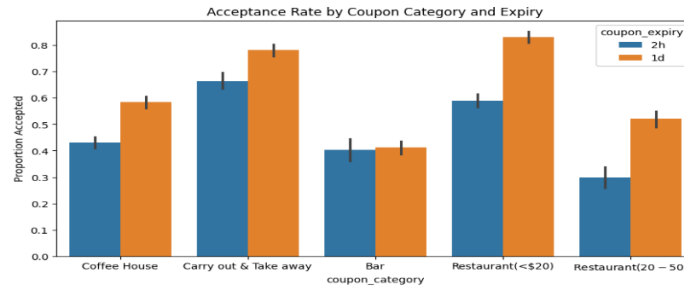


Figure 13: Acceptance Rate by Coupon Type and Expiry

Data Cleaning Issues

Removal of Sparse Columns: Columns with *more than 25% null values* were identified using `spars_cols = df.columns[df.isnull().mean() > 0.25]`. Only *cars*, indicating the driver's mode of transportation, exceeded this threshold. Including it could mislead the model, as imputing so many missing values may create artificial patterns, while dropping rows would reduce dataset size. Therefore, the *cars* column was removed using `.drop()`.

Replacement of Null Values with Mode: Columns *bar*, *coffee_house*, *carry_away*, *restaurant_(<\$20)*, and *restaurant_(\$20-\$50)* had less than 2% null values. These were filled with the **mode** using `.fillna(df[column].mode())`, preserving categorical integrity, since median imputation is not meaningful for nominal data (Ahmed, 2025; ApX, n.d.; Lee, 2025).

Changing Data Types of Certain Columns: The feature age was of 'object' data type and was converted to 'int64' data type for modelling and visualization purposes, it was converted to 'int64' data type. At first, the values in text, below 21 and 50 plus, were replaced by 20 and 55. Then, all the values were converted to integers using the code: `df['age']=pd.to_numeric(df['age'],errors='ignore').astype('Int64')`. Later on, the ages were grouped into ranges for ease of visualization, by using the codes: `conditions[],choices[],df['new_column']=np.select(conditions,choices,default=default_value`.

Correcting Spelling Mistake in Column Names: The column name of "passanger" has been changed to "passenger", thus rectifying the wrong spelling. The Pandas DataFrame (or Series) method used for this purpose was `.rename()`.

Renaming A Few Columns For Clarity: The column names *maritalStatus*, *coupon*, *expiration*, *CoffeeHouse*, *CarryAway*, *RestaurantLessThan20*, *Restaurant20To50*, *toCoupon_GEQ5min*, *toCoupon_GEQ15min*, *toCoupon_GEQ25min*, *Y* were changed for standardization and ease of understanding what they mean as a factor while modelling and visualizations. The Pandas DataFrame method used for this purpose was `.rename()`.

Recoding Inconsistent Categories in Some Categorical Columns: For clarity, response categories in `bar_visits`, `coffee_house_visits`, `carry_away_frequency`, `restaurant_visits_(<$20)`, and `restaurant_visits_($20-$50)` were standardized by replacing never with 0 and gt8 with 8+. Similarly, coupon categories `Restaurant(<20)` and `Restaurant(20-50)` were renamed as `Restaurant(<$20)` and `Restaurant($20-$50)` using `.replace()`.

Rectifying the Chronological Order of Classes in Some Columns: The column `bar_visits` was converted into an ordered categorical variable (e.g., "0" < "1~3" < "4~8" < "8+") using `pd.Categorical()`, and the dataset was sorted with `.sort_values()` to maintain logical order. The same process was applied to `restaurant(<$20)` and `take_out_frequency`. Additionally, categories in temperature, time, age group, income range, and `coupon_expiry` were reordered chronologically using these methods.

Pandas for Insights

The analysis of coupon acceptance across different visit frequencies reveals distinct patterns for various coupon categories. The **acceptance rate of Restaurant (<\$20) coupons** remains consistently high regardless of the frequency of visits to restaurants with expenditures under \$20, indicating strong appeal across all customer segments. Similarly, the **acceptance rate of Carry Out & Take Away coupons** is consistently high across all levels of take-out frequency, suggesting widespread acceptance independent of how often individuals order take-out. In contrast, for other coupon categories—such as Bar, Coffee House, and Restaurant (\$20–\$50)—the **acceptance rate tends to align with the corresponding visit frequency**, meaning that individuals who frequent these establishments more often are more likely to accept coupons for those categories. This indicates that coupon effectiveness for these categories is closely linked to customer behavior and visit patterns.



Figure 16: Acceptance of Takeout Coupons vs Frequency



Figure 17: Restaurant(<\$20) Coupons vs Visits

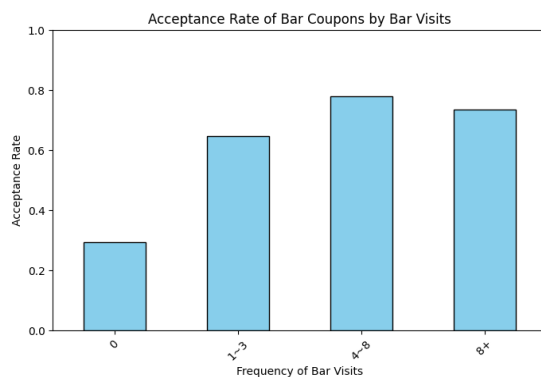


Figure 15: Acceptance Rate of Bar Coupons vs Bar Visits

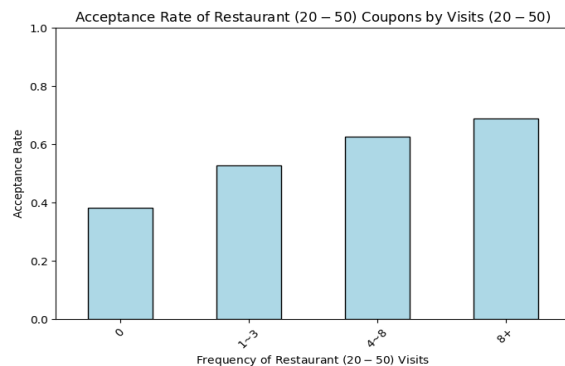


Figure 14: Restaurant(\$20-\$50) Coupons vs Visits

Acceptance of coupon_25 is higher while going in the opposite direction, than while going in the same direction. Divorced males are 79% likely to drive 25 minutes or more to avail an in-vehicle discount coupon, which is highest rate among the people of other marital status of this gender, and **widowed females** are 56% likely to drive 25 minutes or more to avail an in-vehicle discount coupon, which is highest rate among the people of other marital status of this gender.

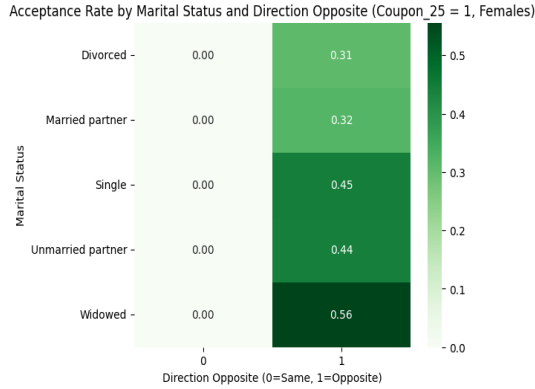


Figure 19: Marital Status vs Direction vs Coupon_25 (Female)

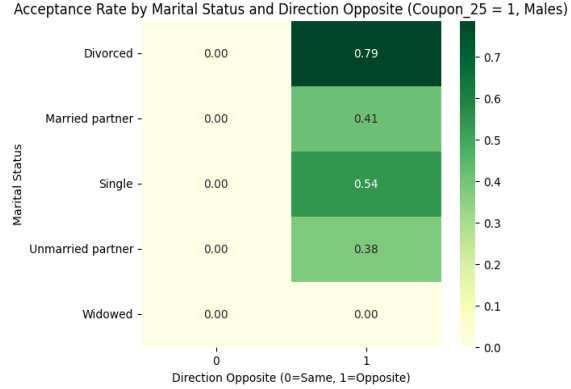


Figure 18: Marital Status vs Direction vs Coupon_25 (Male)

Using several pandas methods, it was found that the **top five occupations with the highest coupon acceptance rates** are Healthcare Support (69.83%), Construction & Extraction (68.83%), Healthcare Practitioners & Technical (67.62%), Protective Service (64.57%), and Architecture & Engineering (63.43%). Surprisingly, these professional groups **show a higher acceptance rate than students**, challenging the common assumption that students are the most likely to accept discount coupons.

Modelling and Performance Comparison

Modelling Context and Rationale

The objective of this analysis is to predict **whether a driver will accept an in-vehicle coupon** under different scenarios. If different food chain businesses can predict which of the drivers will accept the coupons offered, they can concentrate their efforts and resources on those segments. Understanding the factors that influence coupon acceptance can therefore help businesses design targeted promotional strategies, optimize timing and location of offers, and increase customer engagement. By analyzing demographic characteristics, situational factors (such as time, weather, and type of passengers present), and consumer habits, it is aimed to build predictive models that classify drivers as likely or unlikely to accept a coupon. This predictive approach allows companies to implement personalized, data-driven marketing campaigns rather than relying on generic promotions.

Model Description and Modelling Steps

Logistic Regression

Logistic Regression is a linear classification model that predicts the probability of a binary outcome by fitting a logistic function to the input features. It works by estimating the relationship between independent variables and the log-odds of the target event, producing probabilities that can be thresholded to classify observations. In the context of the “In-Vehicle Coupon Recommendation” dataset, logistic regression is used to estimate the probability that a driver will accept a coupon based on features such as their demographic profile (age, gender, income), current driving scenario (destination, time, weather,

passengers), and consumer behavior (frequency of visits to bars, coffeehouses, or restaurants). This allows the model to predict coupon acceptance under different situational and personal contexts, helping businesses identify which drivers are most likely to respond to targeted in-vehicle promotions.

Preprocessing and Encoding:

To build the Logistic Regression model, necessary modules from scikit-learn were imported, including LogisticRegression for modeling, train_test_split for dataset splitting, OneHotEncoder for categorical variables, and metrics such as accuracy_score, confusion_matrix, roc_curve, and roc_auc_score for evaluation.

Input features (X) included demographics, scenario-specific attributes, and consumer habits, while the target variable (y) was accepted_coupon?. All the categorical features were converted to numerical format using **one-hot encoding**, dropping the first category to avoid redundancy. Numerical columns already coded as 0/1 required no scaling. The dataset was split into training and test sets (80-20) with stratification to preserve class distribution. The Logistic Regression model was initialized with the newton-cholesky solver and trained on the training set. Predictions were generated using predict(), and class probabilities using predict_proba() for ROC and AUC analysis.

Support Vector Machine (SVM)

SVM is a classification algorithm that identifies a decision boundary (hyperplane) that best separates classes while maximizing the margin between support vectors, i.e., the points closest to the boundary. By focusing on these critical points, SVM can achieve robust generalization even in complex, high-dimensional spaces. In this dataset, SVM uses features such as driver demographics, trip characteristics, and consumer habits to separate drivers who are likely to accept a coupon from those who are not. The model identifies the optimal hyperplane that distinguishes these two classes, helping marketers understand the key factors that influence coupon acceptance and make predictions for different expected driving scenarios.

Preprocessing and Encoding:

For the SVM model, SVC was imported from scikit-learn (other modules were already available from previous steps). The same input features (X) and target (accepted_coupon?) were used, with categorical variables one-hot encoded and the first category dropped. Numerical features (0/1) required no scaling. The dataset was split into training and test sets (80-20) with stratification.

The SVM was initialized with probability=True to allow ROC-AUC analysis and trained using fit(). Predictions on the test set were generated with predict(), while predict_proba() provided class probabilities for ROC and AUC evaluation.

Decision Tree

Decision Trees classify data by recursively splitting the feature space based on conditions that maximize class separation. Each internal node represents a decision on a feature, and leaf nodes correspond to predicted classes. In the context of the “In-Vehicle Coupon Recommendation” dataset, the decision tree evaluates features like destination urgency, weather, passenger type, coupon category, and driver demographics to make successive decisions that predict coupon acceptance. This approach captures nonlinear relationships and interactions between features, allowing for intuitive interpretation of how different situational and personal factors combine to influence whether a driver accepts a coupon.

Preprocessing and Encoding:

For the Decision Tree model, DecisionTreeClassifier was imported along with plot_tree for visualization (other modules were already imported). Input features (X) and the target variable were the same as before,

with categorical features one-hot encoded and the first category dropped. The dataset was split into training and test sets (80-20) with stratification.

The model was initialized with a maximum depth of 10 to control overfitting and trained using `fit()`. Predictions were generated on the test set with `predict()`, and class probabilities with `predict_proba()` for ROC-AUC analysis. The tree structure was visualized using `plot_tree()` to interpret decision paths and feature importance.

Model Performance Evaluation

All the models were evaluated with accuracy, confusion matrix, precision, recall, F1 score, and ROC-AUC.

Model Name	Accuracy	Precision	Recall (TPR)	F1 Score	ROC-AUC
Logistic Regression	68.78%	70.90%	76.60%	73.60%	73.65%
Support Vector Machine	73.59%	70.90%	76.60%	73.60%	80.54%
Decision Tree	67.90%	72.50%	70.10%	71.30%	71.15%

Table 2: Summary of Performance Metrics

Please refer to Appendix E for Confusion Matrix and ROC curves.

Conclusion

This report analyzed the **In-Vehicle Coupon Recommendation dataset** through cleaning, exploratory analysis, and predictive modeling using Logistic Regression, SVM, and Decision Tree classifiers. The results show that demographic, situational, and behavioral factors can meaningfully predict coupon acceptance, offering practical insights into consumer decision-making. To improve the dataset for future use, reducing missing values, balancing class distributions, and adding richer behavioral attributes would allow for more accurate and robust predictive models.

Lastly, for businesses in Bangladesh, like food delivery, retail, and ride-sharing, such in-vehicle or location-based coupon systems are not yet widely available. Conducting such analysis, along with predictive and classification modeling, can help these sectors identify the right customer segments, design targeted coupon offers, boost redemption rates, and reduce marketing costs, enabling them to effectively introduce and implement in-vehicle coupons in the Bangladeshi market.

References

- Ahmed, S. B. (2025, February 5). *When to Use Mean, Median, and Mode for Handling Missing Values in Data?* <https://www.linkedin.com/pulse/when-use-mean-median-mode-handling-missing-values-data-ahmed-tebje/>
- ApX. (n.d.). *Simple Imputation Methods*. Retrieved August 23, 2025, from <https://apxml.com/courses/intro-feature-engineering/chapter-2-handling-missing-data/simple-imputation>
- Lee, S. (2025, May 15). *Ultimate Guide to Missing Categorical Data*. <https://www.numberanalytics.com/blog/ultimate-missing-categorical-data-guide>

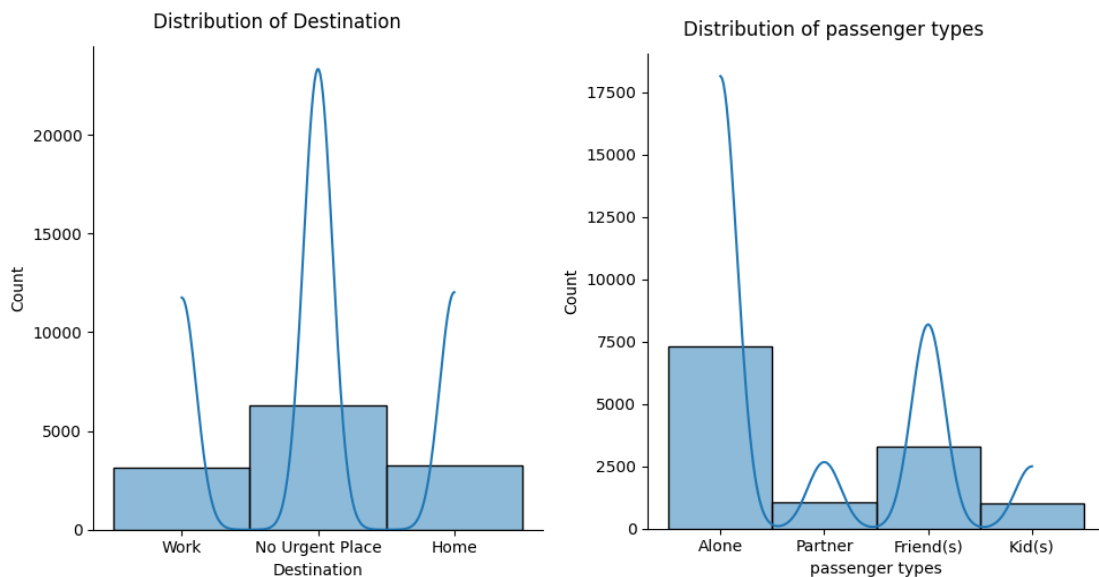
Appendix

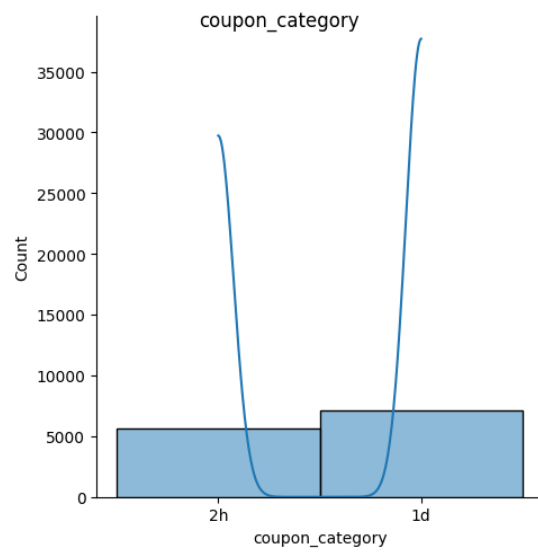
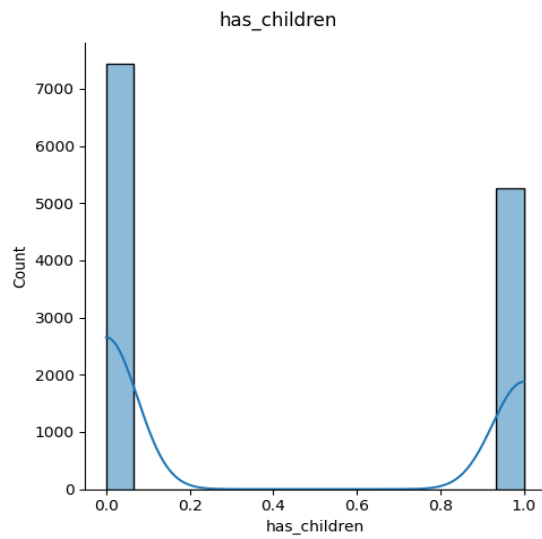
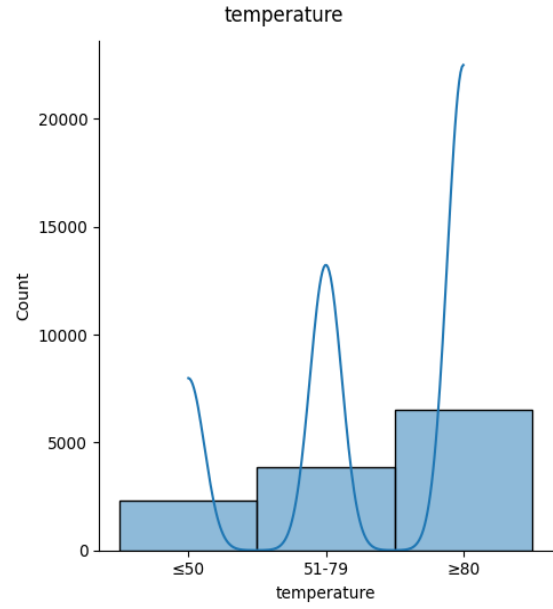
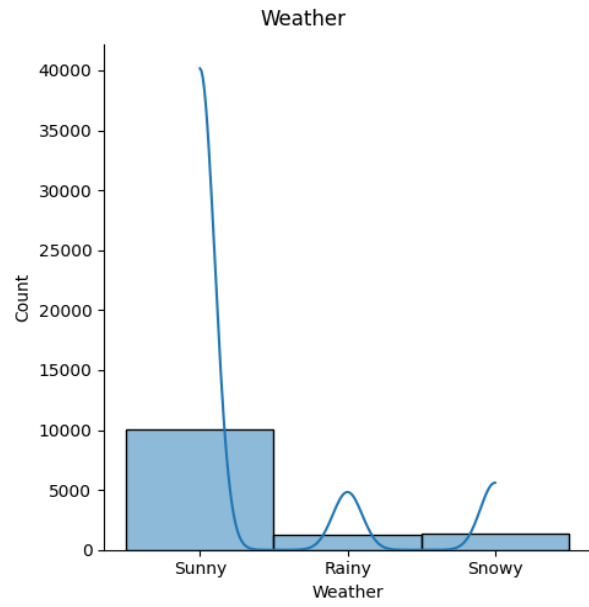
A. Dataset Structure and Feature Overview

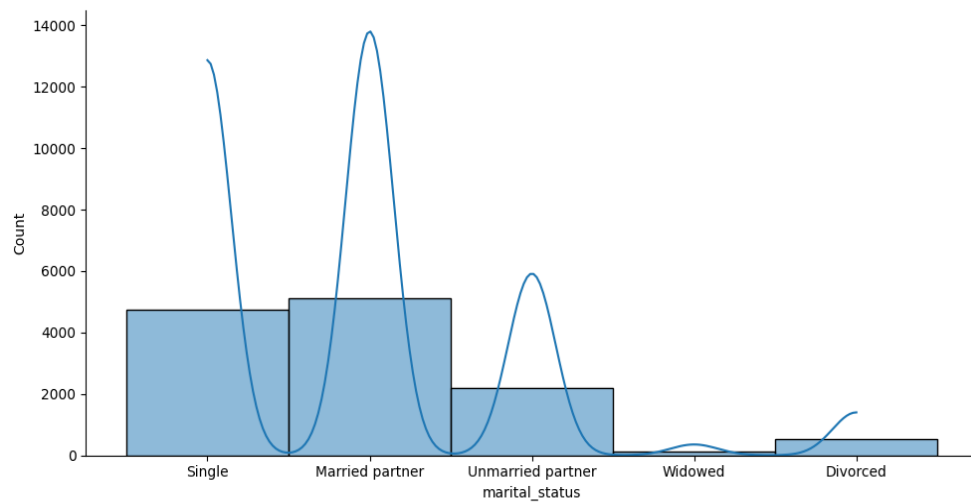
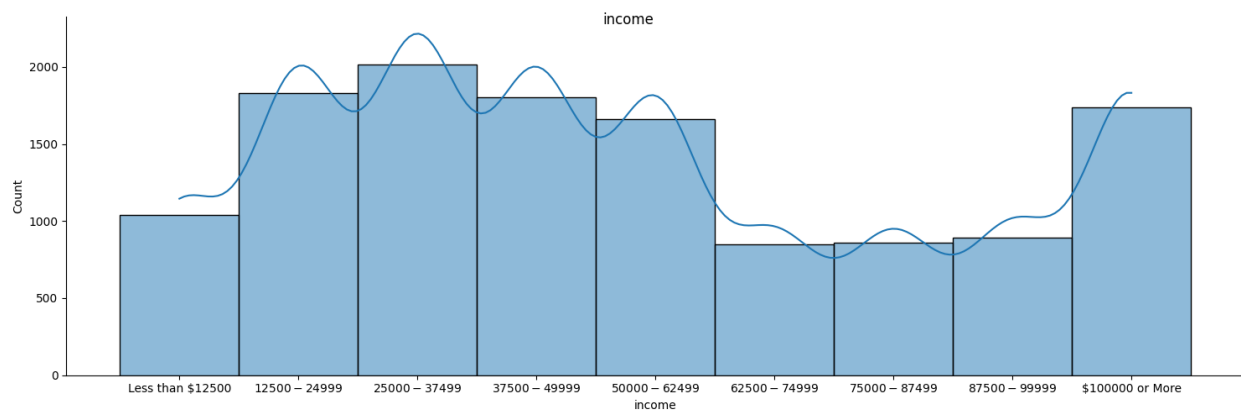
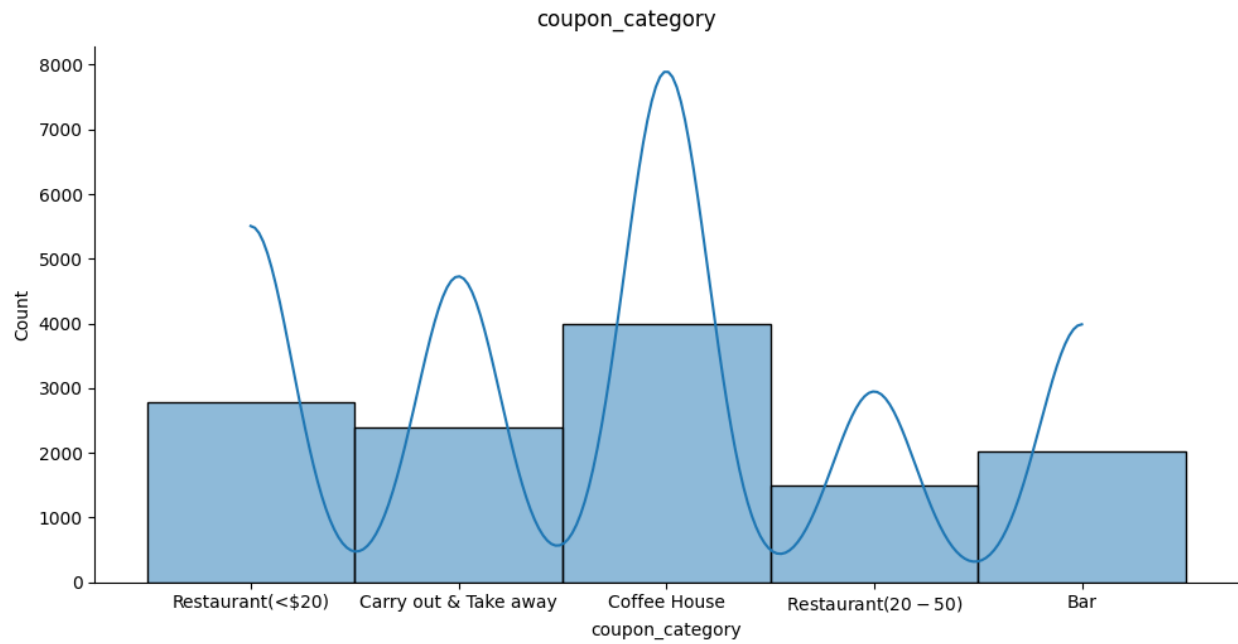
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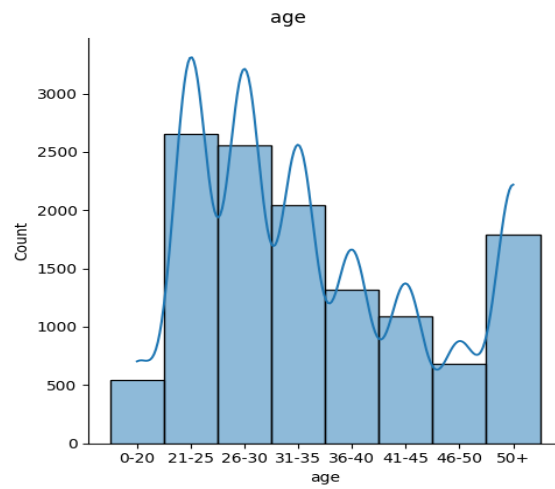
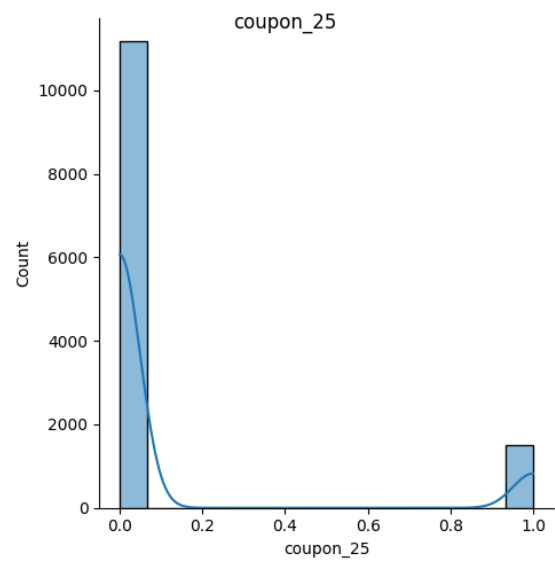
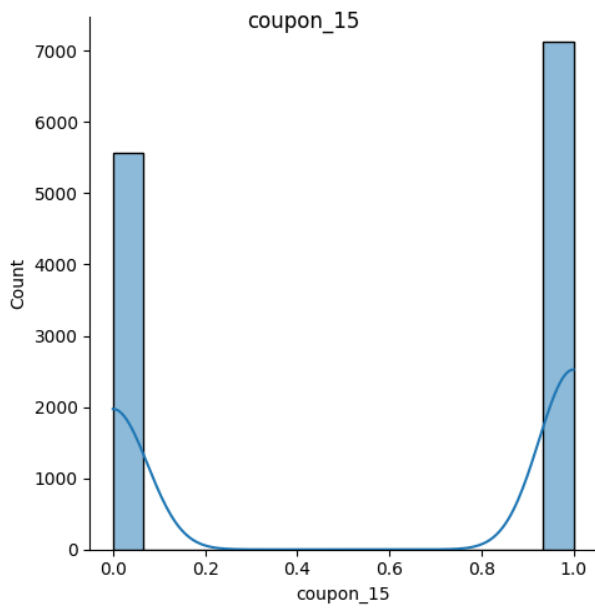
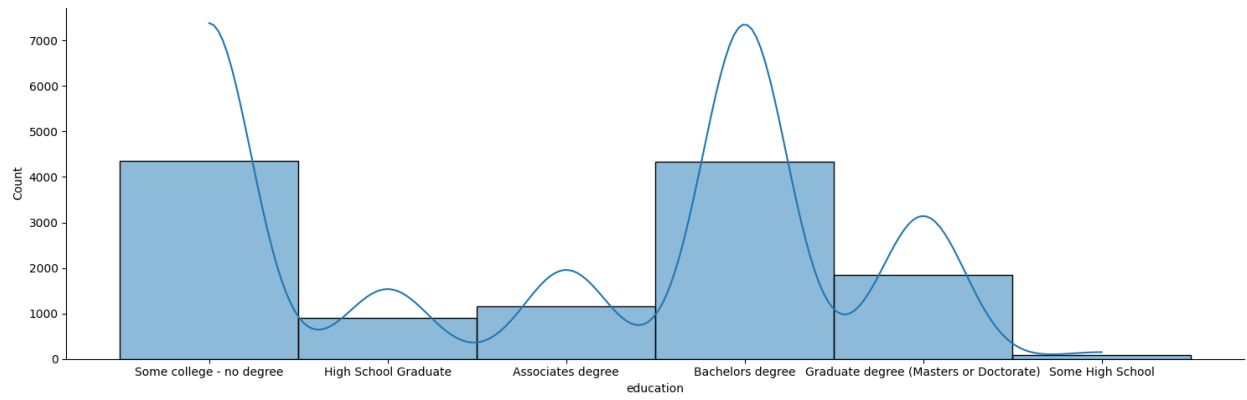
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2   weather                             12684 non-null  object
3   temperature                         12684 non-null  category
4   time                                12684 non-null  category
5   coupon_category                     12684 non-null  object
6   coupon_expiry                      12684 non-null  category
7   gender                              12684 non-null  object
8   age                                 12684 non-null  category
9   marital_status                     12684 non-null  object
10  has_children                        12684 non-null  int64
11  education                           12684 non-null  object
12  occupation                          12684 non-null  object
13  income                             12684 non-null  category
14  bar_visits                          12684 non-null  object
15  coffee_house_visits                 12684 non-null  object
16  take_out_frequency                  12684 non-null  category
17  restaurant_visits_(<$20)            12684 non-null  category
18  restaurant_visits_($20-$50)        12684 non-null  object
19  coupon_5                           12684 non-null  int64
20  coupon_15                          12684 non-null  int64
21  coupon_25                          12684 non-null  int64
22  direction_same                     12684 non-null  int64
23  direction_opp                      12684 non-null  int64
24  accepted_coupon?                   12684 non-null  int64
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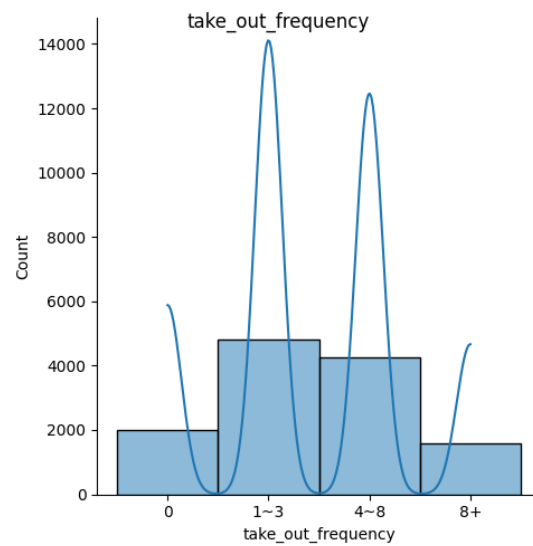
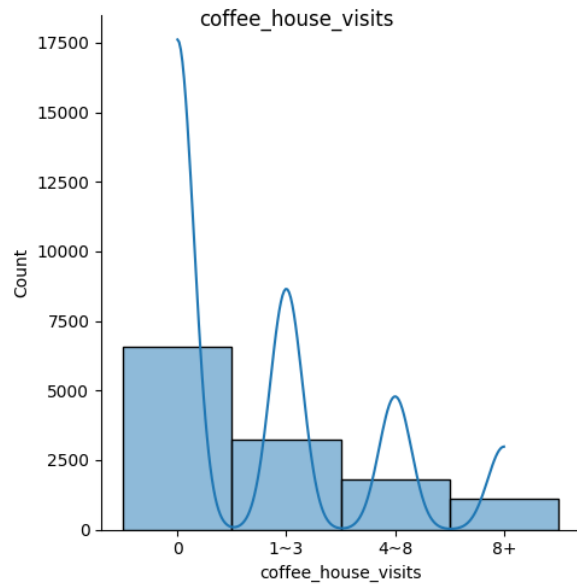
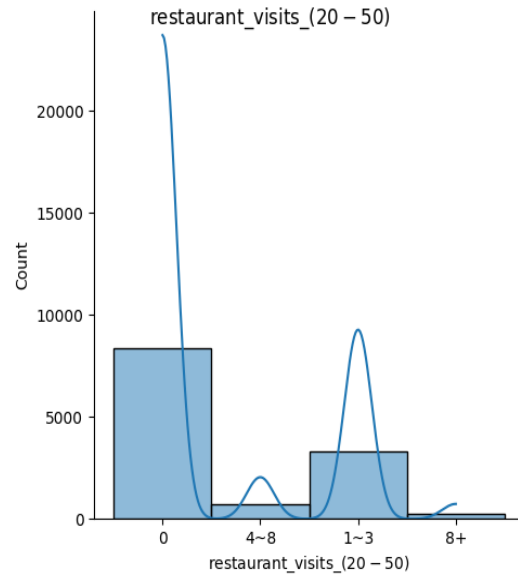
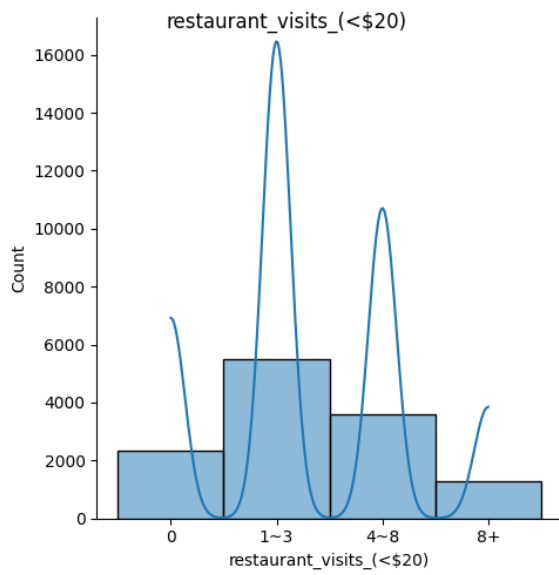
B. Univariate Analyses



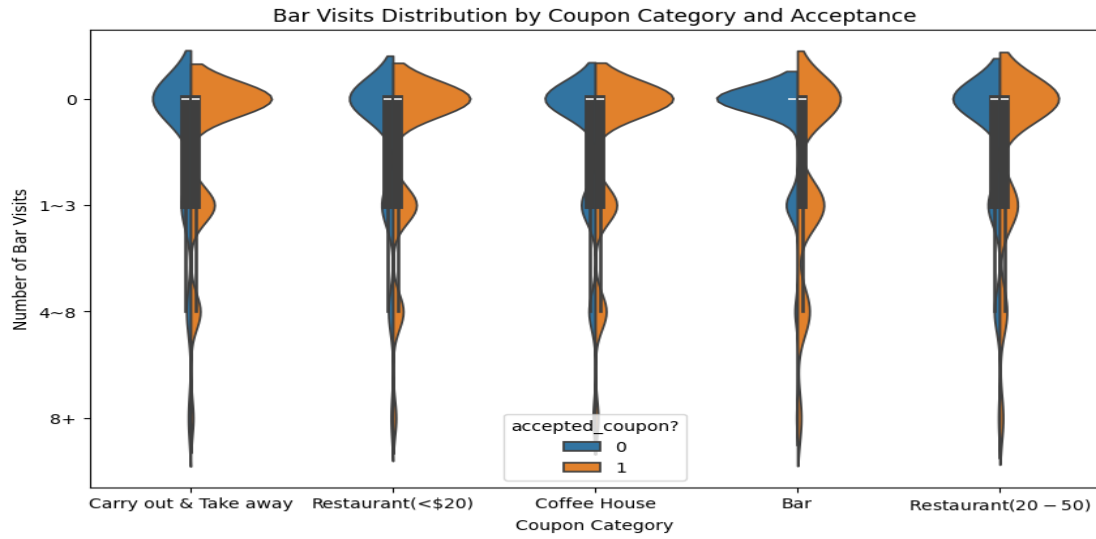




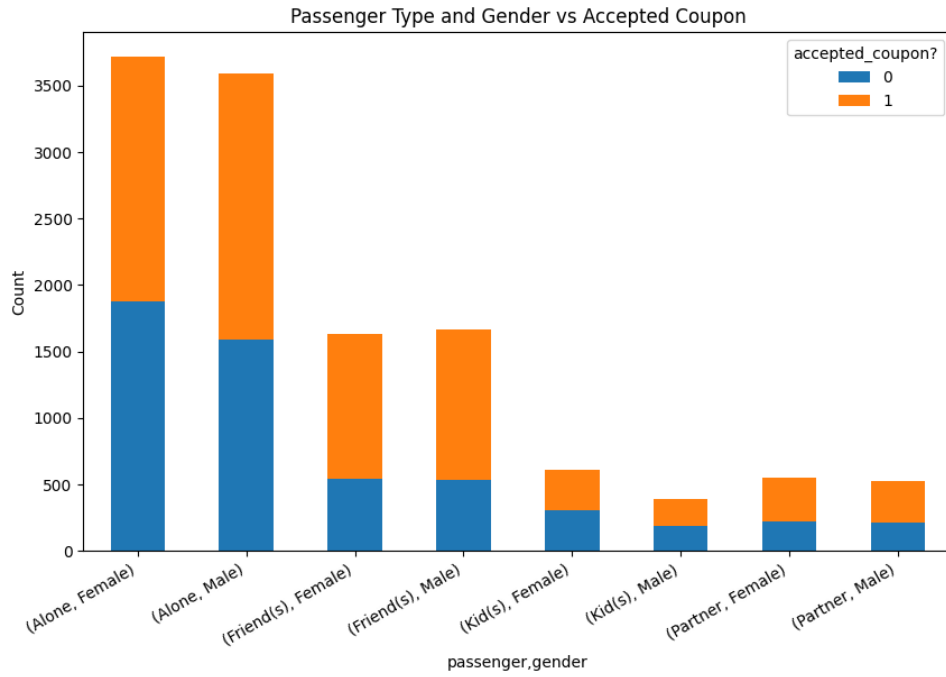


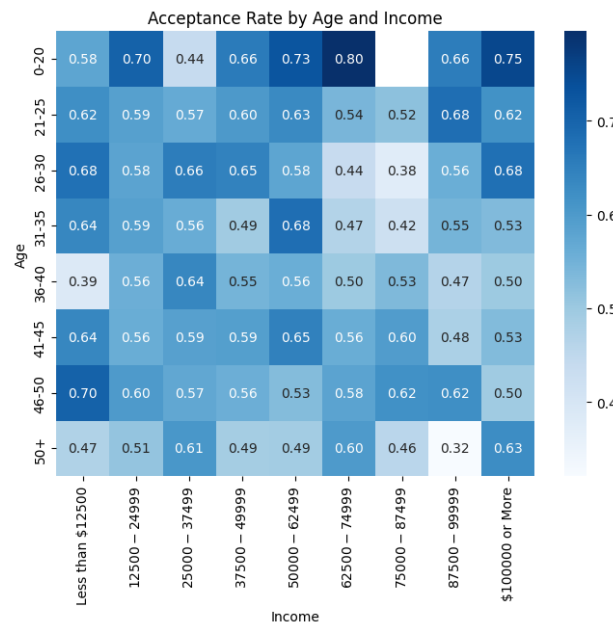
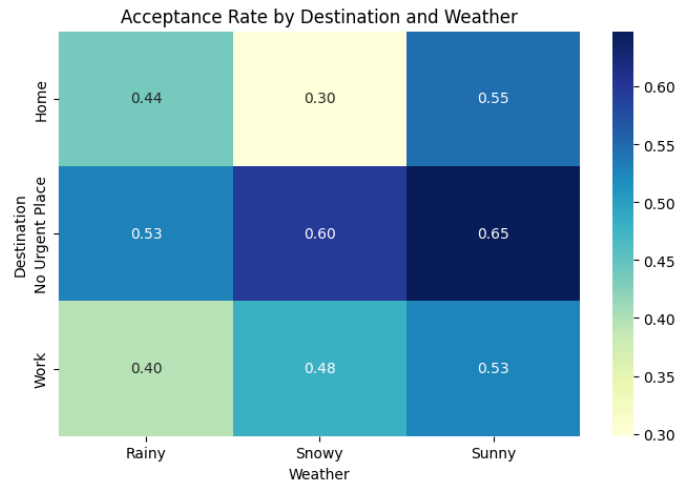


C. Bivariate Analysis



D. Multivariate Analysis





E. Performance Metrics

Confusion matrix of Logistic Regression:

	Model 0	Model 1
Actual 0	641	454
Actual 1	338	1104

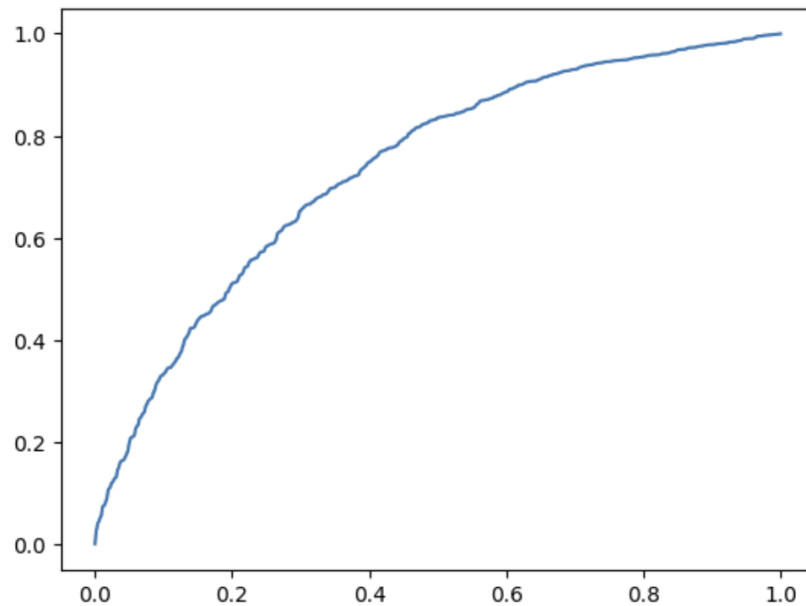
Confusion matrix of SVM

	0	1
0	705	390
1	280	1162

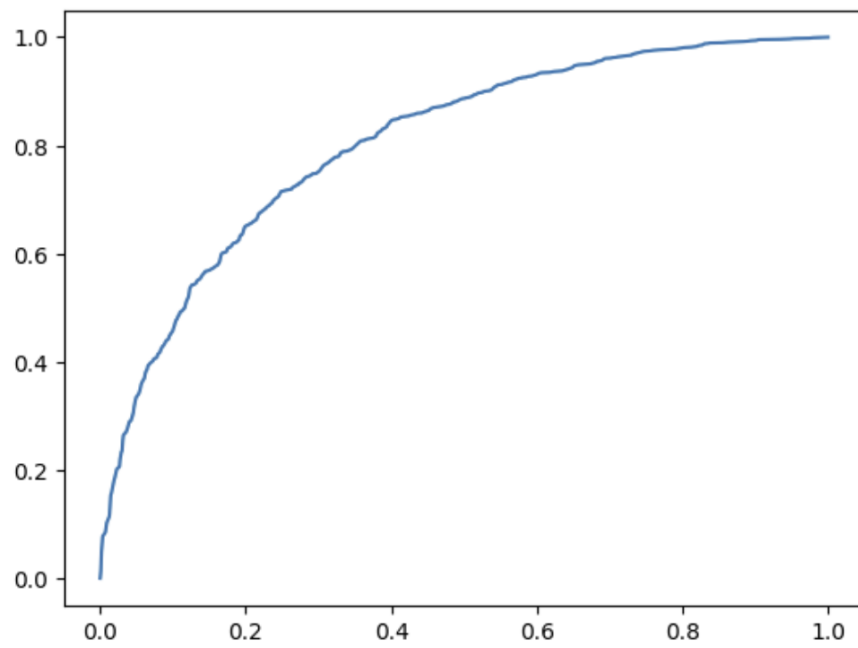
Confusion matrix of Decision Tree

	0	1
0	711	384
1	431	1011

ROC curve of Logistic Regression:



ROC curve of SVM



ROC curve of Decision Tree

